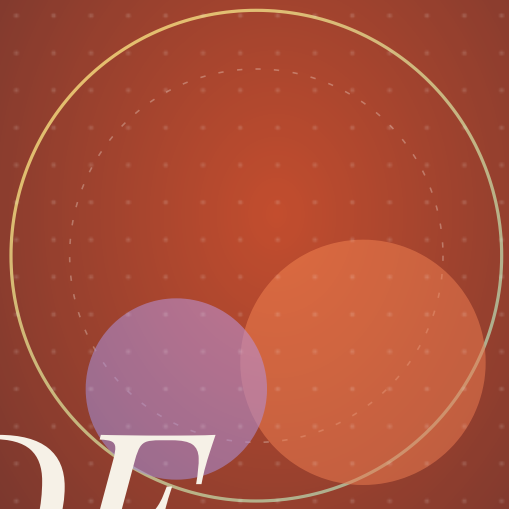
SPECIMEN · VOLUME ONE



CleonPDF Showcase

A tour of the page, glyph by glyph.

Typography, world scripts, vector graphics, mathematics, forms and annotations — gathered in one document to put a PDF engine through its paces.

CLEONPDF
Open-source PDF toolkit in Rust

Generated by `cargo xtask demo-pdf`
US Letter · 8.5 × 11 in

INSIDE THIS SPECIMEN

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COLOPHON

Body text is set in EB Garamond with Playfair Display for display, Inter for labels, and JetBrains Mono for code. Every font is openly licensed (OFL) and listed in ATTRIBUTION.md. Layout is authored in HTML and CSS, printed by headless Chrome, then post-processed with `lopdf`.

WHAT TO LOOK FOR

Open the bookmarks, layers, attachments and comments panels. Page labels run Cover, i–ii, then 1–10. One layer adds a draft watermark; another holds print-only notes and starts hidden.

FOREWORD

Why a *specimen?*

A PDF reader is judged in the first second. The page appears, and either the letters sit on it the way the author intended or they do not. Every kerning pair, every ligature and every soft gradient is a small promise the file makes and the renderer must keep.

This document collects those promises in one place. It was written for the people who build CleonPDF, as a single file to open after every change: if all of it still looks right, the change is probably safe. It is also, we hope, pleasant to read on its own.

Nothing here is exotic. Each feature is something ordinary documents use every day — an invoice with a form field, a thesis full of equations, a contract with comments in the margin, a brochure with a translucent photograph laid over a tinted panel. What is unusual is only that they are all together, and that each page says plainly what it is testing.

The layout is authored in HTML and printed by a browser; the interactive parts are then written into the file by hand, object by object. That split mirrors the real world, where most PDFs are produced by one program and amended by several others.

The CleonPDF contributors

TRACKING, FROM TIGHT TO AIRY

MEASURE TWICE, SET ONCE	-40
MEASURE TWICE, SET ONCE	0
MEASURE TWICE, SET ONCE	+120
MEASURE TWICE	+300

THE FAMILIES IN THIS SPECIMEN

Ag	Ag	Ag	Ag	Ag	∑x
Playfair Display	**EB Garamond**	**Inter**	**JetBrains Mono**	**Great Vibes**	**STIX Two Math**
Display	Body text	Labels	Code	Script	Mathematics

CHAPTER ONE

Setting *Text*

Two justified columns with automatic hyphenation, old-style figures, a drop cap and a pull quote — the everyday machinery of a well-made page.

Long before there were screens, a page of type was a negotiation between the width of a column and the length of the words that had to fill it. The compositor adjusted spaces, divided a word at a sensible syllable, and occasionally rewrote a line so the paragraph would lie flat. Justified text asks for exactly that care, and a renderer must honour the spacing it was given.[1]

A PDF does not reflow. Every glyph has already been placed by the program that wrote the file, and the reader's job is to reproduce those positions faithfully at any zoom. That sounds simple until one remembers that a single line may mix three fonts, a ligature standing in for two letters, and a hyphen that exists only because the line happened to break there in 1994 or 2026.

Old-style figures, like the ones in this paragraph's 1,280 characters and 36 lines, rise and fall with the lowercase so that numbers do not shout. Ligatures such as *fi*, *fl* and *ffi* in words like *officially*, *affluent* and *afflicted* join letters that would otherwise collide.[2]

> *"Good typography is noticed only when it is missing."*
>
> A WORKING MAXIM OF THE TRADE

Hyphenation is where browsers and word processors differ most. Chrome consults the system's hyphenation dictionaries when a paragraph declares its language and asks for automatic hyphens, which is why the words in these columns occasionally divide across lines. The result is a more even texture, with fewer rivers of white running down the column.

Columns introduce their own questions: where to break, whether a heading may sit at the foot of one, and how a quotation spanning both should interrupt the flow. Here the pull quote spans the full measure and the text resumes beneath it, balanced between the two columns.

Small text matters too. Notes, captions and legal lines are often set at seven or eight points, where hinting, stem darkening and subpixel positioning decide whether a reader squints. The footnotes below are set at seven and a half points in the same face as the body.[3]

Finally, notice the drop cap that opens this chapter. It is an ordinary letter, floated and enlarged, with the first line set in small capitals. A renderer need do nothing special for it, which is precisely the point: most typographic richness in a PDF is just careful placement of ordinary glyphs.

1 Word spacing in PDF is usually baked into glyph positions rather than expressed with the word-spacing operator, so readers rarely need to stretch anything themselves.

2 In a PDF a ligature is a single glyph; text extraction relies on the font's ToUnicode map to recover the letters it represents.

3 Try zooming to 400 percent here: well-rendered small text stays crisp and evenly weighted.

CHAPTER TWO

OpenType *Features*

Each row shows the same text with a feature off, then on. The glyphs differ; the underlying characters do not.

TAG	FEATURE	OFF	ON
liga	Standard ligatures	office fluffy fjord	office fluffy fjord
dlig	Discretionary ligatures	Asterisk & strict	Asterisk & strict
smcp	Small capitals	Printed in Boston	Printed in Boston
onum	Old-style figures	Since 1885: 3,467,920	Since 1885: 3,467,920
frac	Fractions	1/2 cup, 3/4 inch, 7/8	½ cup, ¾ inch, ⅞
swsh	Swash capitals	*The Quiet River*	*The Quiet River*
ss01	Stylistic set 1	a 1 4 6 9 G ?	a 1 4 6 9 G ?
kern	Kerning	AVATAR Type Wave	AVATAR Type Wave
tnum	Tabular figures	1111 / 8888	1111 / 8888

ONE VARIABLE FONT, NINE WEIGHTS

Ag 100 Ag 200 Ag 300 Ag 400 Ag 500 Ag 600 Ag 700 Ag 800 Ag 900

System UI · a single variable font file supplies every weight; the PDF embeds each instance as its own subset.

SCRIPT FACES

Handsome letters, joined in flight

Quietly Ornamental — 1234567890

CHAPTER THREE

Expressive *Type*

Effects that become transparency groups, soft masks, clipping paths and transformed text objects once they reach PDF.

Layered

text-shadow · two translucent offset copies composited under the glyphs

Gradient-filled text · an axial shading clipped to glyph outlines (text render mode 7)

Rotated

Skewed

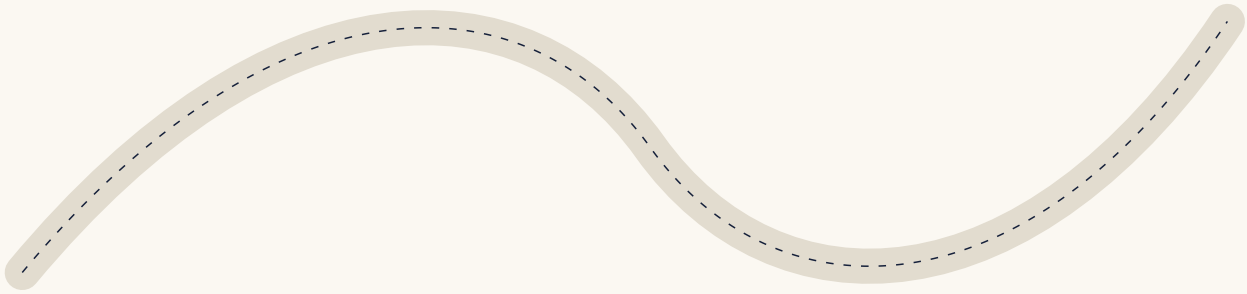
transform · skewX(−18°) on a filled box and rotate(−12°) on a label

Outline

TRACKED TO 600

-webkit-text-stroke · glyphs stroked, not filled (render mode 1)

TEXT ON A CURVE



SVG textPath · each glyph is positioned and rotated individually along a cubic Bézier, filled with a gradient.

HIERARCHY THROUGH CONTRAST

Contrast is the **engine** *of design.*

Pair a high-contrast Playfair italic with a heavy grotesque and the eye knows where to go. The PDF sees only two fonts and a handful of positioning operators; the hierarchy is entirely in the choices.

CHAPTER FOUR

Scripts of the *World*

A short greeting — roughly, "every letter deserves a good home" — in writing systems that need bidirectional layout, contextual shaping, reordering and vertical setting.

ARABIC — RTL · JOINING
لكل حرف بيت
Amiri

HEBREW — RTL
לכל אות יש בית טוב
Noto Sans Hebrew

DEVANAGARI — CONJUNCTS
हर अक्षर का घर
Noto Sans Devanagari

THAI — NO SPACES
ทุกอักษรมีบ้าน
Noto Sans Thai

GREEK — POLYTONIC
Κάθε γράμμα ἔχει σπίτι
EB Garamond

CYRILLIC
Каждой букве — дом
EB Garamond

CHINESE — SIMPLIFIED
每个字都有家
ZCOOL XiaoWei

KOREAN — HANGUL
모든 글자에 집을
Gowun Batang

IPA — PHONETICS
/taɪp ɪz ˈkræft/
International Phonetic Alphabet

ARMENIAN
Ամեն տառին՝ տուն
Noto Sans Armenian

GEORGIAN
ყველა ასოს — სახლი
Noto Sans Georgian

TAMIL
எழுத்துக்கு வீடு
Noto Sans Tamil

JAPANESE
すべての文字に、
よい家を。
縦書きで読む。
Shippori Mincho, writing-mode: vertical-rl

MIXED DIRECTION IN ONE LINE

The word كِتَاب means "book", and ספר does too — the bidi algorithm keeps each run in its own direction while the sentence stays left-to-right, even across numbers like ٣٤٥ and 345.

COLOUR EMOJI

📄 🖋️ 🎨 📐 🔍 🌍 🧱 ✅ 🦀 📎

Noto Color Emoji (OFL) · colour bitmap glyphs that Chrome places into the PDF as images.

CHAPTER FIVE

Mathematics

Equations written in MathML Core and laid out by the browser: stretchy delimiters, radicals, large operators, scripts and matrices.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic formula · fraction, radical, superscript

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Gaussian integral · large operator with limits

$$e^{i\pi} + 1 = 0$$

Euler's identity · five constants, one line

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Basel problem · summation with under- and over-scripts

$$\mathrm{R}\left(\theta\right) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Rotation matrix · the transform behind every rotated page

$$B\left(t\right) = \sum_{i=0}^{3} \binom{3}{i} (1-t)^{3-i} t^i P_i$$

Cubic Bézier · the curve PDF paths are made of

INLINE MATHEMATICS

A PDF page maps user space to device space with a matrix $\left[\,a\ b\ c\ d\ e\ f\,\right]$, so a point $\left(x, y\right)$ lands at $\left(ax + cy + e,\ bx + dy + f\right)$. With one point equal to $\frac{1}{72}$ inch, a Letter page measures 612×792 points, and a 300 dpi raster of it holds roughly 8.4×10^{6} pixels.

SYMBOLS AND ARROWS

$\forall\ \exists\ \varnothing\ \nabla\ \partial\ \in\ \notin\ \subset\ \subseteq\ \cup\ \cap\ \leq\ \geq\ \neq\ \approx\ \equiv\ \infty$

$\aleph\ \mathbb{R}\ \mathbb{C}\ \mathbb{Z}\ \mathbb{N}\ \rightarrow\ \Rightarrow\ \Leftrightarrow\ \mapsto\ \langle\ \rangle\ \lceil\ \rceil$

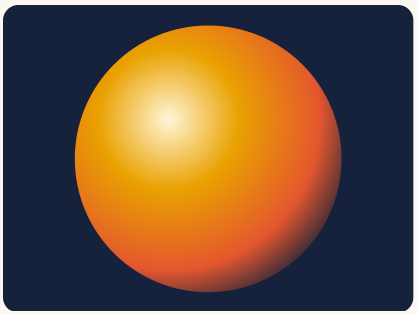
CHAPTER SIX

Vector *Graphics*

Nine small plates, each exercising one piece of the PDF imaging model.



Axial shading · three-stop linear gradient



Radial shading · off-centre focal point



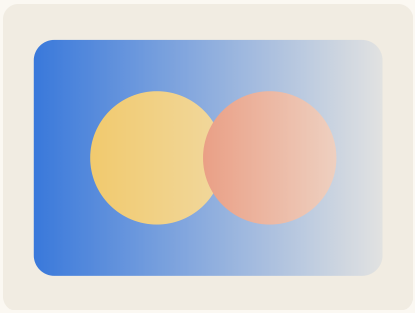
Multiply · subtractive overlaps, /BM /Multiply



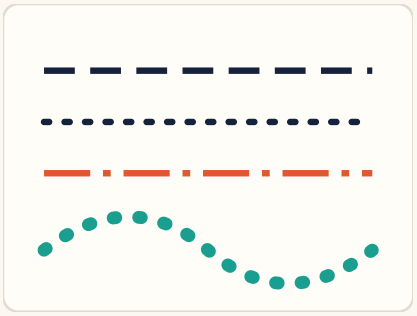
Screen · additive light, /BM /Screen



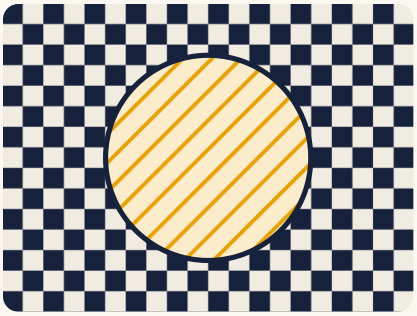
Clipping path · glyph outlines as a clip



Soft mask · luminosity gradient fades artwork



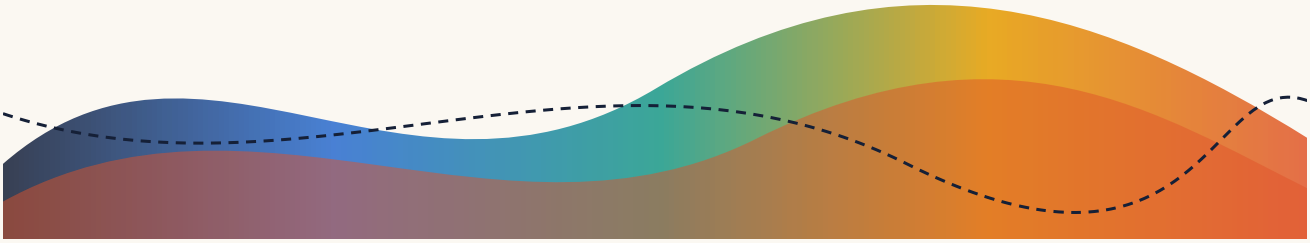
Dash patterns · caps, joins and phase



Tiling patterns · checks and rotated hatch



Fill rules · even-odd pentagram, 45% opacity



Layered waves · five-stop shading, 50% overlay in multiply mode, dashed stroke over the top.

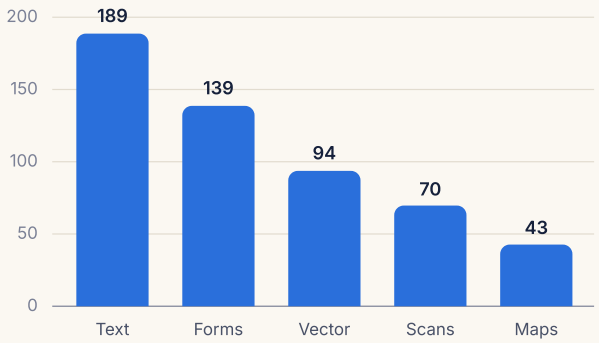
CHAPTER SEVEN

Data & *Tables*

Charts drawn as SVG and a striped table. All figures are illustrative sample data, not measurements.

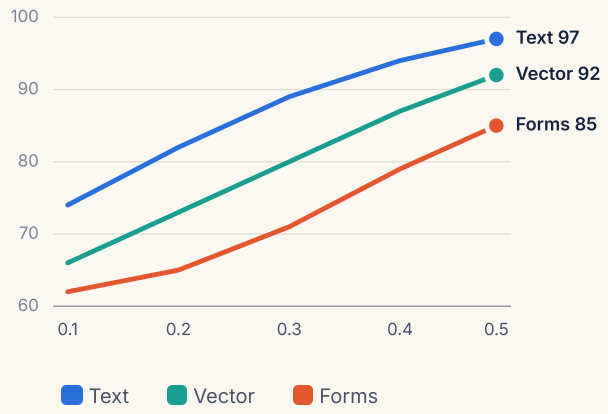
Pages per second, by document type

Illustrative sample data · higher is better



Renderer conformance over releases

Illustrative sample data · % of test pages matching reference



WHERE EACH PDF FEATURE LIVES IN THIS FILE

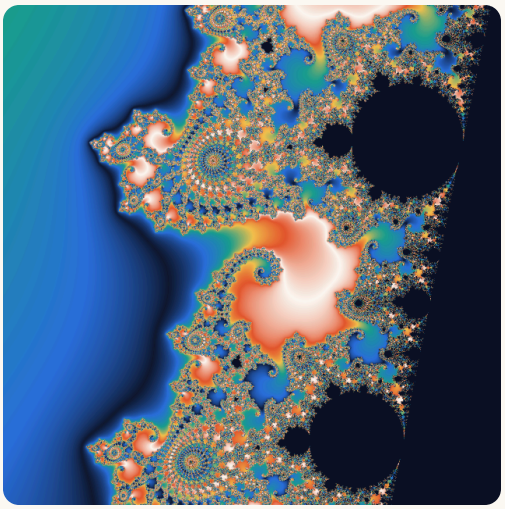
FEATURE	PDF OBJECT	PRODUCED BY	PAGE	STATUS
Embedded font subsets	/FontFile2, /FontFile3	Chrome	all	PRESENT
Tagged structure	/StructTreeRoot	Chrome	all	PRESENT
Axial & radial shadings	/ShadingType 2, 3	Chrome	6	PRESENT
Blend modes	/ExtGState /BM	Chrome	6	PRESENT
Bookmarks	/Outlines	Chrome, xtask	—	PRESENT
Page labels	/PageLabels	xtask	—	PRESENT
Optional content	/OCProperties	xtask	9, 10	TWO LAYERS
Annotations	/Annots with /AP	xtask	1, 9, 10	PRESENT
Form fields	/AcroForm	xtask	10	PRESENT
File attachments	/EmbeddedFiles	xtask	—	PRESENT
Digital signature	/FT /Sig	xtask	10	UNSIGNED

CHAPTER EIGHT

Code & *Images*

A monospaced listing with syntax colouring, and the raster that it describes.

```rust
// Escape-time rendering of Seahorse Valley into an RGB buffer.
fn mandelbrot(width: u32, height: u32) -> Vec<u8> {
    let mut rgb = Vec::with_capacity((width * height * 3) as usize);
    for py in 0..height {
        for px in 0..width {
            let c = (CX + SPAN * (px as f64 / width as f64 - 0.5),
                     CY - SPAN * (py as f64 / height as f64 - 0.5));
            let (mut z, mut n) = ((0.0, 0.0), 0);
            while n < MAX_ITER && z.0 * z.0 + z.1 * z.1 <= 4.0 {
               z = (z.0 * z.0 - z.1 * z.1 + c.0, 2.0 * z.0 * z.1 + c.1);
               n += 1;
            }
            rgb.extend_from_slice(&palette(n)); // smooth colour ramp
        }
    }
    rgb
}
```



A GENERATED RASTER

The image at left is computed by the xtask each time the showcase is built, encoded as PNG and handed to Chrome, which embeds it as an image XObject. Nothing is downloaded and no third-party artwork is involved, so the file stays reproducible and free to redistribute.

At 60 dpi the fractal edge looks smooth; zoom in and the individual pixels appear, a useful check that the viewer switches from filtered to nearest-neighbour sampling at the right moment.

LINKS

- External: https://example.com/
- External: example.org/cleonpdf
- Internal: back to the contents or ahead to review & markup

CHAPTER NINE

Review & *Markup*

Every annotation on this page was added after printing, each with its own appearance stream, author, date and unique name.

TEXT MARKUP

Reviewers rarely rewrite a document; they highlight the phrase that matters, underline a claim that needs a source, and ~~strike out a redundant word~~ before moving on.

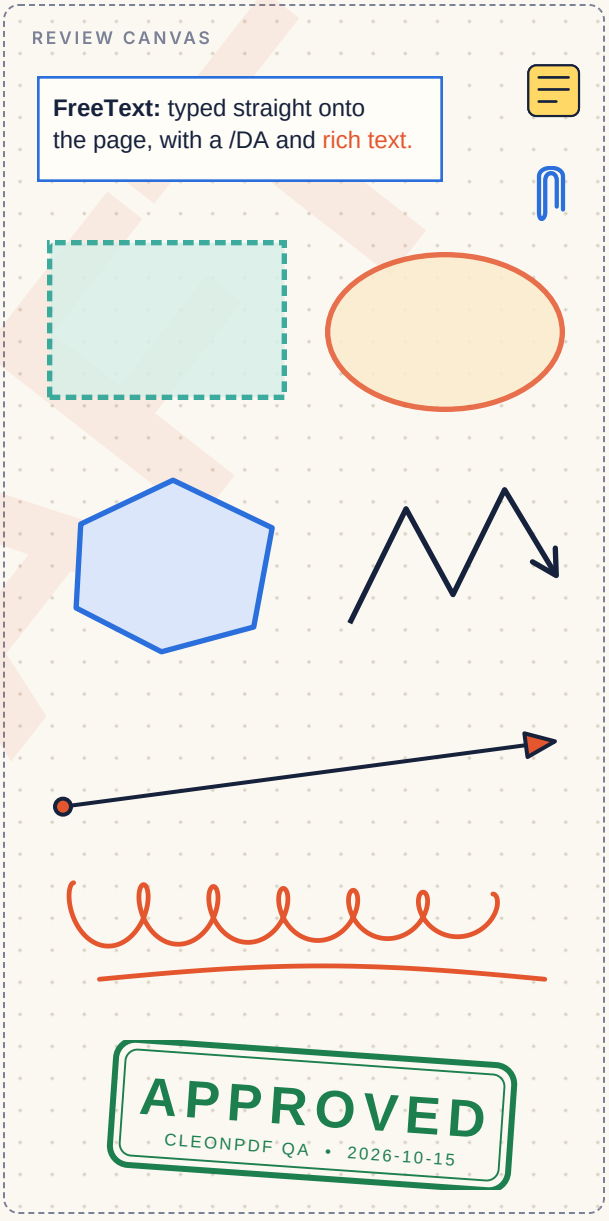
A wavy line says check the spelling here, while a caret marks where text should be inserted.

COMMENTS

Sticky notes carry threaded replies, and a free-text box writes directly on the page. The paperclip holds an attached file that can be saved from the comment itself.

LEGEND

1. Highlight
2. Underline
3. StrikeOut
4. Squiggly
5. Caret
6. Text + Popup
7. FreeText
8. Line (arrow)
9. Square
10. Circle
11. Polygon
12. PolyLine
13. Ink
14. Stamp
15. FileAttachment
16. Link



DRAFT

CHAPTER TEN

Interactive *Form*

This page was not printed by Chrome: the xtask writes its content stream and an AcroForm directly. Every widget ships its own appearance streams.

CONTACT

FULL NAME
Ada Typesetter

EMAIL
ada@example.com

DATE (AFDATE_FORMATEX YYYY-MM-DD)
2026-09-30

AMOUNT (AFNUMBER_FORMAT, RIGHT-ALIGNED)
$1,234.50

COMMENTS (MULTILINE)
CleonPDF renders this field from its own appearance stream, so it looks the same in every viewer. Edit it to see your viewer regenerate the text.

PREFERENCES

CHECKBOXES

- [x] Subscribe to release notes
- [x] I have read the licence
- [ ] Send me stickers

PREFERRED OUTPUT (RADIO GROUP)

- ( ) PDF/A - archival
- (•) PDF/UA - accessible
- ( ) PDF/X - print exchange

PAPER SIZE (COMBO BOX)
Letter

FEATURES OF INTEREST (MULTI-SELECT LIST)
Rendering
Forms
Annotations
Signatures

ACTIONS & SIGNATURE

PUSH BUTTON (JAVASCRIPT ACTION)

Run JavaScript: app.alert

JavaScript: shows app.alert(...) in viewers that run document scripts.

SIGNATURE (UNSIGNED /SIG FIELD)

×

Sign here - unsigned signature field

Links

Back to the contents

https://example.com/